

Disproof of Conjecture 00000003952: twin-width at most 1 is not the same as distance-hereditary

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Abstract

Conjecture 00000003952 asserts, as its first clause, that the graphs of twin-width at most 1 are exactly the distance-hereditary graphs. Both inclusions are false. The house (the complement of the path P_5) has twin-width 1 but is not distance-hereditary. The net (a triangle with a pendant vertex at each corner) is distance-hereditary but has twin-width 2. Since the conjecture is a conjunction, it is false. All definitions (trigraphs, contractions, contraction sequences, twin-width, walks in induced subgraphs, distance-hereditary graphs) are formalized from scratch in Lean 4 (core library only). Both counterexamples are proved there, and `verify.py` checks them independently.

1 The conjecture and the clause we refute

Definition: The twin-width is the minimal width over contraction sequences of red-black graphs. *Conjecture:* The graphs of twin-width at most 1 are exactly the distance-hereditary graphs; recognizing twin-width at most 2 is polynomial time, while recognizing twin-width at most 3 is NP-hard, so the boundary lies exactly at 2.

The Chinese version states the same. The conjecture is a conjunction of three statements. We refute the first one,

$$(C) \quad \{G : \text{tw}(G) \leq 1\} = \{G : G \text{ is distance-hereditary}\},$$

and this already makes the whole conjecture false. We refute *both* inclusions of (C), each with an explicit connected graph. We do not address the two complexity statements.

2 Definitions

All graphs are finite and simple, and have vertex set $\{0, \dots, n-1\}$.

Definition 1 (Trigraphs and contractions; Bonnet–Kim–Thomassé–Watrigant). A *trigraph* (red-black graph) is a vertex set with, for each pair of distinct vertices, one of three colours: *none*, *black* or *red*. A graph is the trigraph with black edges and no red edges. *Contracting* two distinct vertices u, v replaces them by a single new vertex z . Pairs not involving z keep their colours. For every other vertex w , the pair zw is

- black if uw and vw are both black,
- none if uw and vw are both none,

- red otherwise (in particular whenever uw or vw is red).

The *red degree* of a vertex is its number of red edges. A *contraction sequence* of an n -vertex graph G is a sequence $G = T_n, T_{n-1}, \dots, T_1$ of trigraphs, where T_{i-1} is obtained from T_i by contracting two vertices. Its *width* is the maximum red degree of a vertex over all T_i . The *twin-width* $\text{tw}(G)$ is the minimum width over all contraction sequences of G . So $\text{tw}(G) \leq d$ iff some contraction sequence has width at most d .

Equivalently, one can follow the partition of $V(G)$ into the sets of original vertices represented by the current vertices. Two parts $P \neq Q$ are joined by a red edge iff the pair (P, Q) is not homogeneous (some but not all pairs in $P \times Q$ are edges). By induction, this gives exactly the trigraphs above, because the merging rule is the same. `verify.py` uses this partition formulation and Lean uses the trigraph one.

Definition 2 (Distance-hereditary; Howorka). For $S \subseteq V(G)$, $G[S]$ is the induced subgraph and $d_{G[S]}$ its distance (∞ between different components). G is *distance-hereditary* (DH) if $d_{G[S]}(x, y) = d_G(x, y)$ for every S with $G[S]$ connected and all $x, y \in S$, that is, if every connected induced subgraph is isometric.

By Bandelt and Mulder, a connected graph is DH iff it has no induced house, hole C_k ($k \geq 5$), domino or gem. It is also DH iff it can be reduced to a single vertex by repeatedly deleting a pendant vertex or one vertex of a pair of twins. We prove everything from Howorka's definition. The other two characterizations are used only as cross-checks in `verify.py`.

3 The house: twin-width 1, not distance-hereditary

Let H be the graph on $\{0, 1, 2, 3, 4\}$ with edges

$$02, 03, 04, 13, 14, 24.$$

This is the house: the 4-cycle 03140 with the roof vertex 2 adjacent to 0 and 4. Its non-edges $01, 12, 23, 34$ form the path 01234 , so $H = \overline{P_5}$.

Theorem 3. $\text{tw}(H) = 1$, and H is not distance-hereditary.

Proof. $\text{tw}(H) \leq 1$. We contract $\{3, 4\}$, then add 2, then 1, then 0. Here $N(0) = \{2, 3, 4\}$, $N(1) = \{3, 4\}$, $N(2) = \{0, 4\}$, $N(3) = \{0, 1\}$ and $N(4) = \{0, 1, 2\}$.

step	vertices	red edges	max red degree
$T_5 = H$	$0, 1, 2, 3, 4$	none	0
contract $3, 4 \rightarrow A$	$0, 1, 2, A$	$A2$	1
contract $A, 2 \rightarrow B$	$0, 1, B$	$B1$	1
contract $B, 1 \rightarrow C$	$0, C$	$C0$	1
contract $C, 0$	one vertex	none	0

Details:

- A : $3, 4$ are both adjacent to 0 and to 1 (black), and only 4 is adjacent to 2 (red).
- B : $A0$ and 20 are black, so $B0$ is black. $A1$ is black and 21 is none, so $B1$ is red. The red edge $A2$ is now inside B .

- C : $B0$ is black and 10 is none, so $C0$ is red.

Every trigraph has maximum red degree at most 1.

$\text{tw}(H) \geq 1$. Contracting u, v gives the new vertex a red edge to every $w \notin \{u, v\}$ adjacent to exactly one of u, v . If there is no such w , then u, v are twins. H has no twins: the sets $N(x) \setminus \{y\}$ and $N(y) \setminus \{x\}$ above differ for all $x \neq y$, for instance $N(0) \setminus \{4\} = \{2, 3\} \neq \{1, 2\} = N(4) \setminus \{0\}$. So every first contraction creates a red edge, and every contraction sequence has width ≥ 1 .

Not DH. Let $S = \{1, 2, 3, 4\}$. Then $H[S]$ has edges $13, 14, 24$, so it is the path 2413 and is connected. Moreover $d_H(2, 3) = 2$ via 203 , since 2 and 3 are not adjacent. In $H[S]$ the only path from 2 to 3 is 2413 , so $d_{H[S]}(2, 3) = 3 \neq 2$. $\square$

4 The net: distance-hereditary, twin-width 2

Let N be the graph on $\{0, \dots, 5\}$ with edges

$$01, 12, 02, 03, 14, 25.$$

This is the triangle $0, 1, 2$ with pendant vertices $3, 4, 5$ attached to $0, 1, 2$.

Theorem 4. N is distance-hereditary, and $\text{tw}(N) = 2$.

Proof. *DH.* An induced path of N uses at most one triangle edge: if it used two, say abc , then ac would be a chord. A pendant vertex has degree 1, so it can only be an endpoint of a path. So an induced path from x to y consists of, in order:

- the edge from x to its triangle vertex, if x is a pendant;
- the triangle edge between the two triangle vertices, if they differ;
- the edge to y , if y is a pendant.

Hence all induced x - y paths of N have the same length. A shortest path is induced, so this common length is $d_N(x, y)$. Now let $N[S]$ be connected and $x, y \in S$. A shortest x - y path in $N[S]$ is induced in $N[S]$, hence induced in N , so its length is $d_N(x, y)$. Therefore $d_{N[S]}(x, y) = d_N(x, y)$. (Alternatively: delete the pendants $3, 4, 5$; the remaining triangle reduces by twin deletions.)

$\text{tw}(N) \geq 2$. By the remark in the proof of Theorem 3, the new vertex gets a red edge to every vertex of $\Delta(u, v) = (N(u) \setminus \{v\}) \Delta (N(v) \setminus \{u\})$. Up to the symmetry of N , which permutes the three corners together with their pendants, there are four types of pairs:

$$\Delta(0, 1) = \{3, 4\}, \quad \Delta(0, 3) = \{1, 2\}, \quad \Delta(0, 4) = \{2, 3\}, \quad \Delta(3, 4) = \{0, 1\}.$$

These are triangle-triangle, corner with its own pendant, corner with another pendant, and pendant-pendant. In every case the first contraction already creates a vertex of red degree 2. So no contraction sequence has width ≤ 1 .

$\text{tw}(N) \leq 2$. Contract 3 into 0 , 4 into 1 , 5 into 2 , then 1 into 0 , then 2 into 0 . The maximum red degrees along this sequence are $0, 2, 2, 2, 1, 0$. $\square$

Theorem 5. *Clause (C) is false in both directions: H has $\text{tw} \leq 1$ but is not DH, and N is DH but has $\text{tw} > 1$. Hence Conjecture 00000003952 is false.*

5 Interpretation and robustness

- **“Exactly”** means equality of the two graph classes. Each inclusion fails separately (Theorems 3 and 4).
- **Twin-width.** We use the standard definition. The trigraph and partition formulations agree, as explained above. “Red-black graphs” are the trigraphs, and a graph is a trigraph without red edges. The house sequence has red degree ≤ 1 at *every* vertex of every trigraph. The net lower bound comes from the red degree of the *newly created* vertex in the first trigraph. So both results also hold for variants that measure only the contracted vertex, or all vertices. Twin-width does not depend on vertex names, and in Lean the merged vertex keeps the name of the first one.
- **Distance-hereditary.** Both graphs are connected, so the usual definitions coincide on them: Howorka’s isometric-subgraph definition, the Bandelt–Mulder forbidden-subgraph and pruning characterizations, and the condition that every induced path is a shortest path. The house is itself one of the forbidden induced subgraphs.
- **Context.** Twin-width is invariant under complementation, but the DH property is not: $\overline{H} = P_5$ is DH. This explains why (C) cannot hold. `verify.py` also enumerates all connected graphs on at most 6 vertices:
 - on 5 vertices, exactly two have $\text{tw} \leq 1$ but are not DH (the house and the gem);
 - on 6 vertices there are 27 such graphs, and the net is the only DH graph with $\text{tw} > 1$.
- **Not addressed.** The complexity statements about recognizing twin-width at most 2 and at most 3 are not needed and not discussed.

6 Lean formalization

The directory `lean4/` is a self-contained Lean 4.19.0 project (core library only; no `sorry`, no `native_decide`, no added axioms; `#print axioms` reports at most `propext` and `Quot.sound`). Contents of `Main.lean` (namespace `TwinWidth`):

- `Graph` (vertex count and edge list), `Graph.adj`, `Graph.WF` (well-formed simple graph).
- `Col`, `Trigraph`, `Trigraph.contract`: the contraction rule above, via `mergeCol`. Also `redDeg`, `maxRedDeg`, `Graph.toTrigraph` (all edges black).
- `IsContractionSeq`: every step contracts two distinct current vertices, and at most one vertex remains at the end.
- `trigraphsAlong` and `seqWidth`: the maximum red degree over all trigraphs of the sequence.
- `TwwLe G d` := $\exists s$, s is a contraction sequence of G with `seqWidth` $\leq d$. `TwwEq G d` means `TwwLe G d` and not `TwwLe G d'` for $d' < d$.
- `Reach G m k x y`: an inductive walk predicate. It says there is a walk $x = v_0, \dots, v_j = y$ with $j \leq k$ and $v_1, \dots, v_j$ in the vertex set encoded by the bit mask m , i.e. $d_{G[m]}(x, y) \leq k$.
- `ConnectedOn G m`, and `DH G`: for all masks $m < 2^n$ with $G[m]$ connected, all $x, y \in m$ and all k , `Reach G (full G) k x y` implies `Reach G m k x y`.

- Computation:
 - `reach_iff_ball` proves that `Reach` agrees with a computable BFS ball.
 - `dh_of_check` proves, for every well-formed graph, that a finite Boolean check `dhCheck` implies DH. The check includes stabilization of the balls at radius n .
 - `not_twwLe_of_first` proves that if every first contraction has red degree $> d$, then `TwwLe G d` fails.
- Results:
 - `house_twwLe_one`, `house_seq_widths` (red degrees $[0, 1, 1, 1, 0]$), `house_tww_eq_one`.
 - `house_not_DH`, with witness $m = \{1, 2, 3, 4\}$, $x = 2$, $y = 3$, $k = 2$.
 - `net_DH`, `net_not_twwLe_one`, `net_tww_eq_two`.
 - Sanity checks: `P5_twwLe_one` and `P5_DH`. These show that both predicates are satisfiable and agree on P_5 .
- Final theorems:
 - `twwToDH_false` : $\sim \text{ClauseTwwToDH}$
 - `DHToTww_false` : $\sim \text{ClauseDHToTww}$
 - `conjecture_00000003952_false` : $\sim \text{Clause}$, where
 $\text{Clause} := \forall G, G.WF \rightarrow (\text{TwwLe } G \ 1 \leftrightarrow \text{DH } G)$
 - `conjecture_00000003952_conjunction_false` (Rest) : $\sim (\text{Clause} \wedge \text{Rest})$

Lean does not cover the enumeration of all graphs on at most 6 vertices or the equivalence with other DH characterizations. These appear only in `verify.py` and are not needed for the disproof.

7 Independent check and reproduction

`verify.py` uses only the Python standard library (plus `nauty-geng` for the optional enumeration, if installed). It checks:

1. the explicit sequences, by trigraph simulation;
2. the exact twin-width of H , N and P_5 , by exhaustive search over partition sequences;
3. the DH property, in three ways: BFS distances in all induced subgraphs, Bandelt–Mulder pruning, and forbidden induced subgraphs;
4. the enumeration mentioned above.

```
cd lean4 && lake build      # Lean 4.19.0, core library only
cd .. && python3 verify.py
```

References

- [1] É. Bonnet, E. J. Kim, S. Thomassé, R. Watrigant, *Twin-width I: tractable FO model checking*, J. ACM 69 (2022).
- [2] E. Howorka, *A characterization of distance-hereditary graphs*, Quart. J. Math. Oxford 28 (1977) 417–420.
- [3] H.-J. Bandelt, H. M. Mulder, *Distance-hereditary graphs*, J. Combin. Theory Ser. B 41 (1986) 182–208.